

LINKING CALIFORNIAN CLIMATE VARIABILITY TO GRAPE YIELD LOSSES WITH INSURANCE PREMIUMS

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CONTEXT

- WHY THIS MATTERS

**CALIFORNIA PRODUCES WORLD'S MOST
VALUABLE WINES**

**CLIMATE THREATS INTENSIFYING:
DROUGHT, WILDFIRE SMOKE**

ECONOMIC STAKES

\$48B production value 1999-2020)
in our 16-county sample

QUESTION

Does crop insurance protect
growers as climate extremes
worsen?



METHODOLOGY

- OUR APPROACH

Climate

- Temperature
- Precipitation
- Drought indices

Production

- Production
- Value
- Yield

Insurance

- Premium
- Indemnities
- Loss Ratios

Panel Data

- 16 counties
- Top 82% Producer
- 22 years

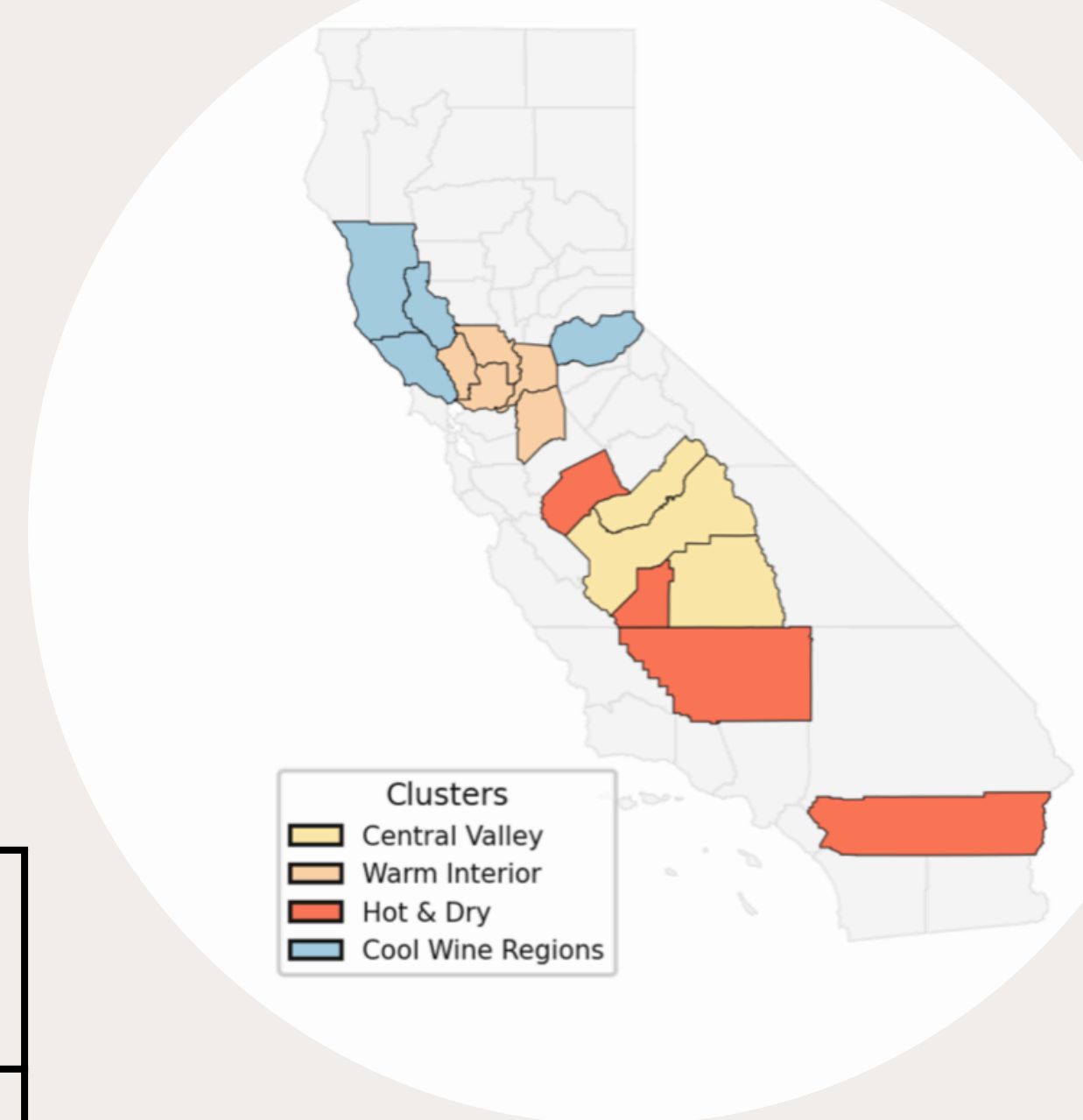
Three Analyses

- Climate Clustering → 4 risk zones
- Drought Impact (2012-2016)
- Insurance Vulnerability



FINDINGS #1
– CLIMATE IS NOT UNIFORM

Cluster	Production VS Temperature	Risk Level
Hot & Dry	-0.68	Critical
Central Valley	+0.63	Moderate
Warm Interior	+0.36	Low
Cool Wine Regions	+0.58	Low



FINDINGS #2

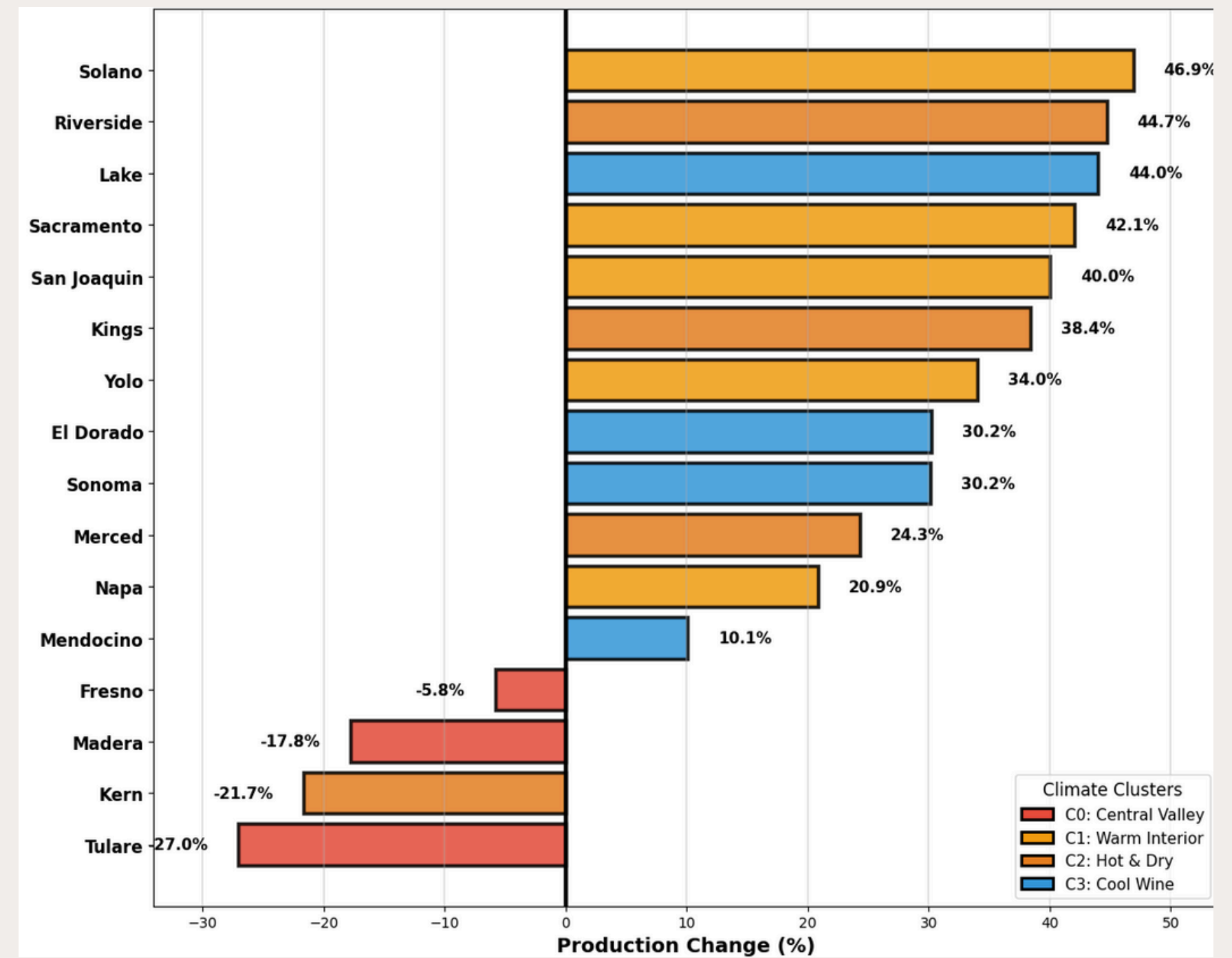
– THE DROUGHT PARADOX

DECLINING *Tulare Basin*

- Groundwater only
 - Over-pumped aquifers
- Wells **failed**

GROWING *Rivers/Aqueducts*

- Surface water access
 - River irrigation
- Adapted **successfully**





FINDINGS #3

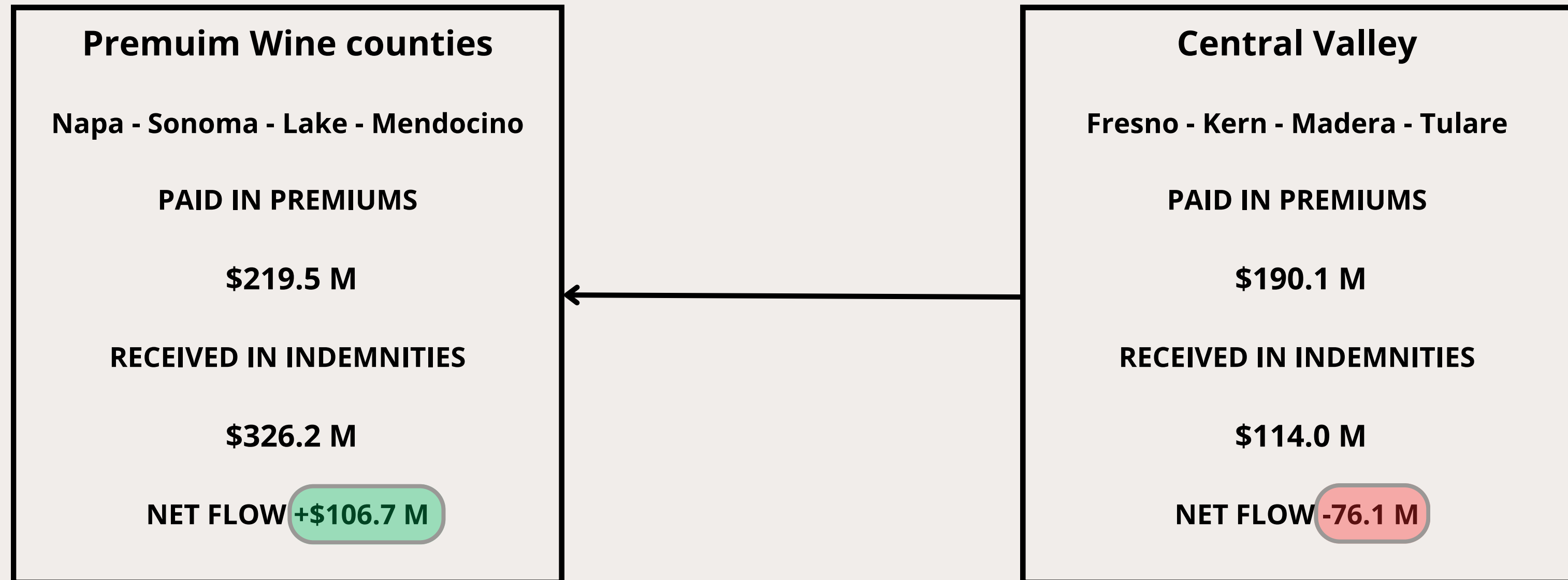
– THE INSURANCE GAP

Region Type	County	Value/Acre	Coverage	Uninsured	Loss ratio
Premium Wine	Napa	\$12,982	22%	9.2B	1.11
	Sonoma	\$8,400	29%	7.1B	1.08
Central Valley	Fresno	\$3,104	55%	2.2B	0.66
	Madera	\$2,587	43%	1.6B	0.44

THE PARADOX: HIGHEST VALUE = LOWEST PROTECTION

FINDINGS #4

– WHO PAYS?



CONCLUSION

KEY FINDINGS:

Geography > Climate alone:

Water infrastructure determines drought vulnerability, not rainfall.

Risk is concentrated:

Hot, dry counties appears closer to climate limits while others benefit from modest warming.

Insurance misalignment:

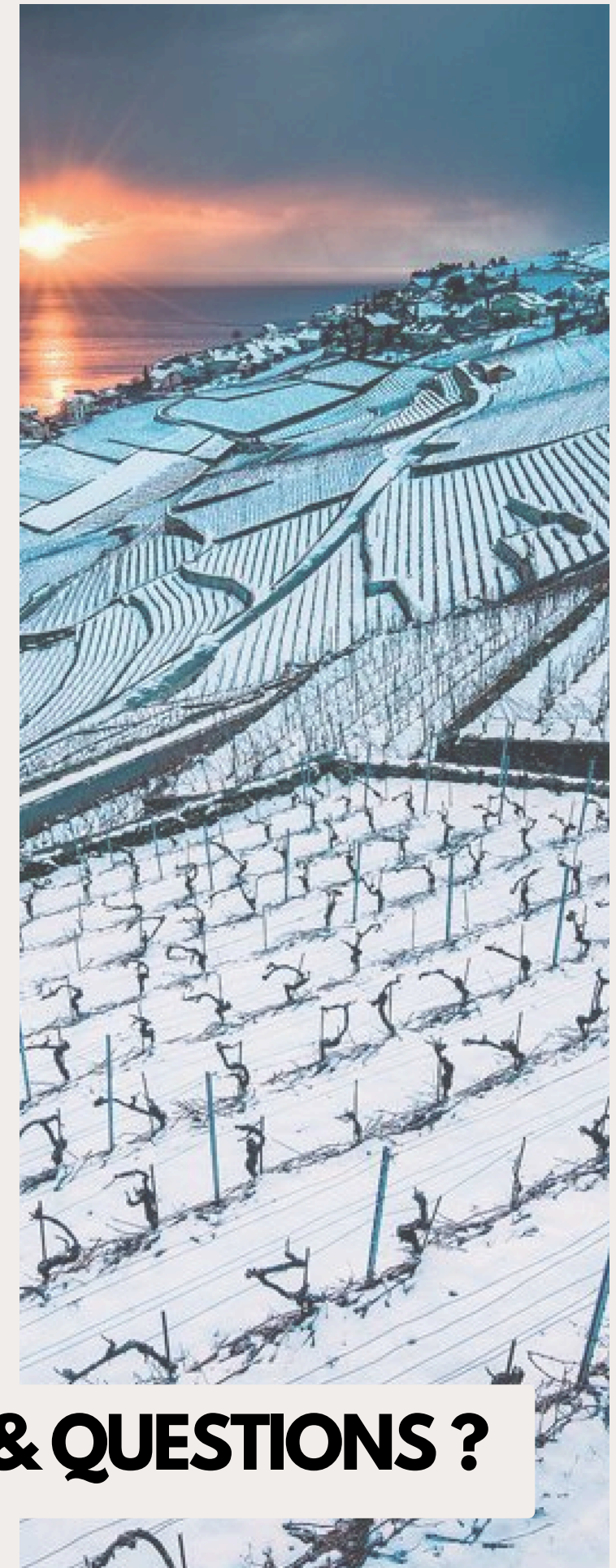
Highest-value regions have lowest coverage and cross-subsidize from lower-value producers.

POLICY IMPLICATIONS:

- ✓ Need region-specific pricing reflecting water access, not just weather.
- ✓ Current system leaves \$16B+ uninsured exposure in most vulnerable regions.
- ✓ As climate extremes intensify, this gap threatens California's premium wine industry.

FUTURE RESEARCH:

Integrate wildfire/smoke data (2017-2020), AVA-level analysis, quality vs. quantity losses



THANK YOU & QUESTIONS ?